The tangent groupoid, its zoom, and whole trajectories

Correction, 2026-10-27. The earlier numbered summary below and corresponding passages are historical and their broad identifications are withdrawn. The groupoid zoom is a geometric rescaling; identifying it with integration of fluctuations or with a running physical action constant requires additional arguments. A real-time Hamilton principal function composes by stationary elimination on an appropriate branch, not by an unconditional global minimum. The separate claim that minimization is an ordinary groupoid-morphism property was also false: §3 replaces it by a min-plus kernel semigroup and gives a counterexample. Euclidean Brownian scaling is not a probability law for real-time Feynman paths, and an unqualified claim that quantum paths have Hausdorff dimension two is not justified. The precise state-level construction, the different scales and a Gaussian scaling test are given in the later tangent-groupoid formulation. The constant-force calculation remains a separately scoped example.

Result, 2026-09-27 (user's question: what Connes's tangent groupoid adds, beyond the user's 1998 and 2003 readings, and whether it extends to a whole trajectory). Three statements.

1. **The zoom is the renormalization.** Debord and Skandalis's action of $\mathbb{R}_+^*$ on the tangent groupoid, $(x, y, \varepsilon) \mapsto (x, y, \lambda^{-1}\varepsilon)$ and $(x, v, 0) \mapsto (x, \lambda v, 0)$, is the dilation family of the fifth-postulate note (§4, Theorem D) and the control transformation $h \mapsto \mu h$ of the user's 1998 note. By van Erp and Yuncken, pseudodifferential operators are exactly the distributions on the tangent groupoid that are homogeneous under the zoom up to smoothing terms; their restriction to $\varepsilon = 0$ is the principal symbol. What survives the passage $\varepsilon \to 0$ is therefore the principal symbol, a function of place and velocity together.
2. **Trajectories live on a time-graded groupoid.** The tangent groupoid composes chords at one fixed ε, and at $\varepsilon = 0$ it composes only vectors at one point. Refining a trajectory into more pieces is an operation on the groupoid $(x, y, u) \circ (y, z, v) = (x, z, u + v)$ that the user introduced in 2003 as the "elementary school" groupoid. When a classical action actually minimizes, its endpoint kernels satisfy $\inf_y [S_r(x, y) + S_v(y, z)] = S_{r+v}(x, z)$, a semigroup law under min-plus convolution. This holds for Galileo's constant force in Proposition 1 below; general real-time principal actions require stationary elimination on a specified branch. Quantum kernels compose by integration, interpreted as operator composition when necessary. A Euclidean small-$\hbar$ Laplace limit can relate the two kernel laws under additional hypotheses.
3. **Along a quantum trajectory the ultimate velocity fails.** Quantum paths have Hausdorff dimension two: $\langle \Delta x^2 \rangle \sim \hbar\tau/M$ over a step τ, so $\Delta x/\tau$ diverges like $\tau^{-1/2}$. The tangent groupoid's $\varepsilon \to 0$ fibre, where $(y - x)/\varepsilon \to v$, is reached by ballistic scaling; quantum trajectories need the diffusive scaling $\Delta x^2 \propto \tau$, the user's 1998 remark that the path groupoid relates to Connes's only after $(x, y, \varepsilon) \mapsto (x, y, \varepsilon^2)$, and his 2003 remark on the $\sqrt{t}$ of Itô's calculus. Newton's *velocitas ultima* is the ballistic boundary fibre, which quantum paths do not reach.

The geometric constructions are established; the broader identifications are historical readings subject to the correction above. Proposition 1 is the separately scoped constant-force kernel calculation.

1. Connes's construction

For a manifold M, the tangent groupoid is $\mathbb{G}_M = (M \times M \times (0, 1]) \cup (TM \times \{0\})$, with the pair-groupoid law at each $\varepsilon > 0$, the fibrewise vector addition $(x, X) \circ (x, Y) = (x, X + Y)$ at $\varepsilon = 0$, and the topology in which $(x_n, y_n, \varepsilon_n) \to (x, v, 0)$ iff $x_n \to x$ and $(y_n - x_n)/\varepsilon_n \to v$ (in a chart). Its C^*-algebra is a continuous field whose fibre at $\varepsilon > 0$ is the compact operators on $L^2(M)$ and at $\varepsilon = 0$ is $C_0(T^*M)$ by Fourier transform; Connes used the deformation to prove the

index theorem (A. Connes, *Noncommutative Geometry*, Academic Press 1994, §II.5, metadata), and the construction generalizes the Moyal rule (Cariñena, Clemente-Gallardo, Follana, Gracia-Bondía, Rivero and Várilly 1999, abstract; Landsman 2002, arXiv:math-ph/0208004, abstract). The $\varepsilon = 0$ fibre is where a chord has become a velocity at a place: Newton's *velocitas ultima*, "certain and definite", with place and velocity commuting. The fibre at $\varepsilon > 0$ is noncommutative.

2. The zoom action and what survives

Debord and Skandalis (2014) (metadata) define the action of $\mathbb{R}_+^*$ on the adiabatic (tangent) groupoid displayed in the summary and identify the crossed product with an ideal of pseudodifferential operators; van Erp and Yuncken (2019) (metadata) characterize pseudodifferential operators as the distributions on the tangent groupoid, properly supported and transversal, that are homogeneous under the zoom modulo smooth ones. Three identifications follow (readings):

- the zoom on the $\varepsilon = 0$ fibre, $v \mapsto \lambda v$, is dual on T^*M to the scaling that carries the Moyal product with constant $\hbar$ to the one with constant $\lambda^{\pm 2}\hbar$ (fifth-postulate note, §4);
- the user's 1998 "dressed series", invariant under $h \mapsto \mu h$ (arXiv:quant-ph/9803035, eq. (10)), is a zoom-invariance condition, and his Ehrenfest remark (eq. (11)) is the statement that the equations of motion are zoom-invariant (Theorem A of the fifth-postulate note);
- the principal symbol, the $\varepsilon = 0$ restriction of a zoom-homogeneous distribution, is the classical datum that survives the limit. The atlas's question "what survives refinement" has here a precise answer for one step.

3. Whole trajectories: the time-graded groupoid

The user's 2003 note (Rivero, "Flashes of noncommutativity", arXiv:math/0302285, full read) introduces the groupoid over configuration space with arrows (x, y, u), u an elapsed time, and law $(x, y, u) \circ (y, z, v) = (x, z, u + v)$. Its convolution algebra has product $(AB)(x, z, t) = \iint A(x, y, r)\, B(y, z, t - r)\, dy\, dr$; Fourier transformation in t gives, with $\varepsilon \sim 1/\hat{t}$, the product of the $\varepsilon > 0$ part of the tangent groupoid, so the latter's algebra sits inside the former's (the 2003 note's second flash). Refining a trajectory is factoring an arrow (x, z, u) through intermediate points and times, which is Newton's insertion of $t_{2.6}$.

For endpoint functions define min-plus convolution by

$$(A \otimes B)(x, z) := \inf_y \{A(x, y) + B(y, z)\}.$$

It eliminates the intermediate endpoint; it does not add the action on each individual composable pair of arrows.

Proposition 1. For $M > 0$, constant force F, and $u > 0$, set

$$S_u(x, y) = \frac{M(y - x)^2}{2u} + \frac{Fu}{2}(x + y) - \frac{F^2 u^3}{24M}.$$

Then $(S_u)_{u>0}$ is a semigroup of kernels under min-plus convolution: $S_r \otimes S_v = S_{r+v}$ for $r, v > 0$. This is the corrected action ℓ_u of the refinement note, equation (2). The corresponding constant-force unitary propagators satisfy $U_{r+v} = U_r U_v$; their kernels compose by integration over the intermediate position, in the oscillatory/distributional sense where the real-time integrals are not absolutely convergent.

Proof. The sum $S_r(x, y) + S_v(y, z)$ is a strictly convex quadratic in y with unique minimizer

$$y_* = \frac{vx + rz}{r + v} - \frac{Frv}{2M}.$$

Substitution gives $S_{r+v}(x, z)$. Equivalently, y_* is the value at time r of the action-minimizing constant-force path joining x to z. The unitary law follows from the self-adjoint Hamiltonian; its kernel statement is operator composition. □

The min-plus law is not an ordinary groupoid morphism. For free motion with $M = 1$, $S_1(0,1) + S_1(1,0) = 1$, whereas $S_2(0,0) = 0$. The equality holds only after minimizing over the intermediate point ($y = 0$ in this example).

The displayed S is the exact discrete Lagrangian of discrete mechanics (Marsden and West 2001, metadata), and discrete Lagrangian mechanics on groupoids is Weinstein's (1996) (metadata). The kick–drift–kick cell $S_h = K_{Fh/2} D_h K_{Fh/2}$ of the refinement note is the variational (Störmer–Verlet) integrator of the trapezoidal discrete Lagrangian L_h, and the cubic counterterm turns L_h into the exact one. The laws of Proposition 1 can be related through a Euclidean Laplace limit: under coercivity and regularity conditions, $-\hbar \log \int e^{-S/\hbar} dy \to \inf_y S$ as $\hbar \to 0$ (Litvinov 2005, metadata). This does not turn an oscillatory real-time integral into a probability integral, nor make every stationary real-time branch a global minimum.

4. Quantum trajectories miss the ultimate velocity

In a path integral the typical increment over a step τ has $\langle \Delta x^2 \rangle \sim \hbar\tau/M$, so the paths have Hausdorff dimension two (Abbott and Wise 1981, metadata), and the difference quotient $\Delta x/\tau \sim (\hbar/M\tau)^{1/2}$ diverges. The tangent groupoid reaches its $\varepsilon = 0$ fibre along ballistic sequences, $(y - x)/\varepsilon \to v$. Quantum trajectories approach along diffusive ones, $\Delta x^2 \propto \tau$: the user's 1998 remark that Connes's groupoid relates to the groupoid of paths only after the rescaling $(x, y, \varepsilon) \mapsto (x, y, \varepsilon^2)$, and his 2003 remark on the $\sqrt{t}$ of Itô's calculus, are this fact. The dimensionless ratio $M\,\Delta x^2/(\hbar\tau)$ is the action per step in units of $\hbar$: ballistic refinement sends it to zero (the classical limit, Newton's ultimate velocity), diffusive refinement holds it fixed (the dimension ladder: refining makes each Newtonian cell quantum). The failure of joint determinacy of the fifth-postulate note is, along a trajectory, the failure of the ballistic limit.

5. Consequence for STATE

The tangent groupoid organizes geometric deformation, and trajectories can be factored in a time-graded arrow space. For the constant-force example, refinement gives a min-plus kernel semigroup classically and operator composition quantum mechanically; neither is an ordinary additive morphism of individual arrows. Identifying the geometric zoom with a particular RG integration remains a separate task. See the later state formulation for the controlled quantum/classical comparison.